Exercise-1

Marked Questions can be used as Revision Questions.

OBJECTIVE QUESTIONS

Section (A) : Energy Band

A-1. The forbidden energy band gap in conductors, semiconductors and insulators are EG_1, EG_2 and EG_3 respectively, The relation among them is

(1) $EG_1 = EG_2 = EG_3$ (2) $EG_1 < EG_2 < EG_3$

(3) $EG_1 > EG_2 > EG_3$ (4) $EG_1 < EG_2 > EG_3$

A-2. Fermi energy is the

(1) Minimum energy of electrons in metal at 0 K

(2) Minimum energy of electrons in metal at 0°C

(3) Maximum energy of electrons in metal at 0 K

(4) Maximum energy of electrons in metal at 0°C

A-3. Which of the following statements is true

(1) In insulators the conduction band is completely empty.

(2) In conductor the conduction band is completely empty.

(3) In semiconductor the conduction band is partially empty at low temperature.

(4) In insulators the conduction band is completely filled with electrons.

A-4. Which of the following statement is true-

(1) At absolute zero temperature, the semiconductor behave as a conductor

(2) The energy gap in semiconductor is more than that for insulator

(3) The resistance of semiconductor increases with increase in temperature

(4) The resistance of semiconductor decreases with increase in temperature

A-5. The valence band at (0 K) is-

(1) completely filled (2) completely empty (3) partially filled (4) nothing can be said

A-6. The free electron concentration (n) in the conduction band of a semiconductor at a temperature T Kelvin is described in terms of E_g and T as –

(1) $n = ATe^{-Eg/kT}$ (2) $n = AT^2e^{-Eg/kT}$ (3) $n = AT^2e^{-Eg/2kT}$ (4) $n = AT^{3/2}e^{-Eg/2kT}$

A-7. Electric conduction in a semiconductor takes place due to

(1) electrons only (2) holes only

(3) both electrons and holes (4) neither electron nor holes

A-8. An electric field is applied to a semiconductor. Let the number of charge carries be n and the average drift speed be v. If the temperature is increased,

(1) both n and v will increase (2) n will increase but v will decrease

(3) v will increases but n will decrease (4) both n and v will decrease

A-9. The mobility of free electron is greater than that of free holes because

(1) They carry negative charge (2) They are light

(3) They mutually collide less (4) They require low energy to continue their motion

Section (B) : Intrinsic semiconductor

B-1. Let n_p and n_e be the numbers of holes and conduction electrons in an intrinsic semiconductor
(1) $n_p > n_e$ (2) $n_p = n_e$ (3) $n_p < n_e$ (4) $n_p \neq n_e$

B-2. The electrical conductivity of pure germanium can be increased by
(1) increassing the temperature (2) doping acceptor impurities
(3) doping donor impurities (4) irradiating ultraviolet light on it.

Section (C) : Extrinsic semiconductor

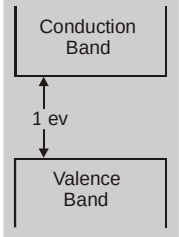
C-1. A N-type semiconductor is
(1) Negatively charged (2) Positively charged (3) Neutral (4) None of these

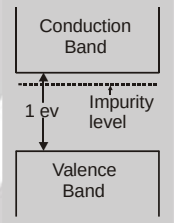
C-2. Let n_p and n_e be the numbers of holes and conduction electrons in an extrinsic semiconductor
(1) $n_p > n_e$ (2) $n_p = n_e$ (3) $n_p < n_e$ (4) $n_p \neq n_e$

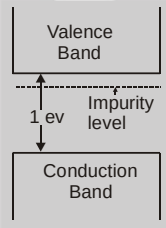
C-3. A semiconductor is doped with a donor impurity
(1) The hole concentration increases (2) The hole concentration decreases
(3) The electron concentration increases (4) The electron concentration decreases

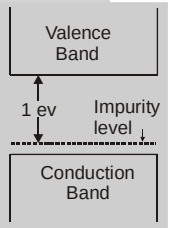
C-4. In a P-type semiconductor, the acceptor level is 57 meV, above the valence band. The maximum wave length of light required to produce a hole will be-
(1) 57 A^o (2) 57×10^{-3} A^o (3) 217100 A^o (4) 11.61×10 A^o

C-5. Which of the following energy band diagram shows the N-type semiconductor-

(1) 

(2) 

(3) 

(4) 

C-6. A P-type silicon semiconductor is made by adding one atom of indium per 5×10^7 atoms of silicon is 25×10^{28} atom/m.^{3}. Point the number of acceptor atoms in per cubic cm. of silicon
(1) 2×10^{30} atom/cm^3 (2) 5×10^{15} atom/cm^3
(3) 1×10^{15} atom/cm^3 (4) 2.5×10^{36} atom/cm^3

C-7. If N-type semiconductor is heated then-
(1) The number of electrons increases and the number of holes decreases
(2) The number of holes increases and the number of electrons decreases
(3) The number of electrons and holes both remain equal
(4) The number of both electrons and holes increases equally

C-8. GaAs is-
(1) an elemental semiconductor (2) a compound semiconductor
(3) an insulator (4) a metallic semiconductor

C-9. What will be conductance of pure silicon crystal at 300K temp.? If electron hole pairs per cm^3 is 1.072×10^{10} at this temp., $\mu_n = 1350$ cm^2/volt sec and $\mu_p = 480$ cm^2/volt sec-
(1) 3.14×10^{-6} mho/cm (2) 3×10^6 mho/cm
(3) 10^{-6} mho/cm (4) 10^6 mho/cm

C-10. Forbidden energy gap of Ge is 0.75 eV, maximum wave length of incident radiation for producing electron-hole pair in germanium semiconductor is-
(1) 4200 A^o (2) 16500 A^o (3) 4700 A^o (4) 4000 A^o

C-11. Mobility of electrron in N-type Ge is 5000 cm^2 / volt sec and conductivity 5 mho/cm. If effect of holes is negligible then impurity concentration will be-
(1) $6.25 \times 10^{15}/cm^3$ (2) $9.25 \times 10^{14}/cm^3$ (3) $6 \times 10^{13}/cm^3$ (4) 9×10^{13} / cm^3

C-12. The intrinsic carrier density in germanium crystal at 300 K is 2.5×10^{13} per cm^3. If the electron density in an N-type germanium crystal at 300 K be 0.5×10^{17} per cm^3, the hole density (per cm^3) in this N-type crystal at 300 K would be expected around-
(1) 2.5×10^{13} (2) 5×10^6 (3) 1.25×10^{10} (4) 0.2×10^4

C-13. Pure Si at 300 K has equal electron (n_e) and hole (n_h) concentrations of 1.5×10^{16} m^{-3}. Doping by indium increases n_h to 4.5×10^{22} m^{-3}. Calculate n_e in the doped Si-
(1) 5.0×10^9 m^{-3} (2) 6.0×10^6 m^{-3} (3) 7.0×10^3 m^{-3} (4) 4.0×10^9 m^{-3}

C-14. The length of a germanium rod is 0.58 cm and its area of cross-section is $1mm^2$. If for germanium $n_i = 2.5 \times 10^{19}$ m^{-3}, $\mu_h = 0.19$ m^2/ V-s, $\mu_e = 0.39$ m^2/ V-s, then the resistance of the rod will be -
(1) 2.5 KΩ (2) 5.0 KΩ (3) 7.5 KΩ (4) 10.0 KΩ

C-15. The contributions in the total current flowing through a semiconductor due to electrons and holes are $\frac{3}{4}$ and $\frac{1}{4}$ respectively. If the drift velocity of electrons is $\frac{5}{2}$ times that of holes at this temperature, then the ratio of the concentrations of electrons and holes is-
(1) 6 : 5 (2) 5 : 6 (3) 3 : 2 (4) 2 : 3

Section (D) : Diodes

D-1. The P-N junction is-
(1) an ohmic resistance (2) an non ohmic resistance
(3) a positive resistance (4) a negative resistance

D-2. Diffusion current in a p-n junction is greater than the drift current in magnitude
(1) if the junction is forward-biased (2) if the junction is reverse-biased
(3) if the junction is unbiased (4) in no case

D-3. A hole diffuses from the p-side to the n-side in a p-n junction. This means that
(1) a bond is broken on the n-side and the electron freed from the bond jumps to the conduction band
(2) a conduction electron on the p-side jumps to a broken bond to complete it
(3) a bond is broken on the n-side and the electron freed from the bond jumps to a broken bond on the p-side to complete it
(4) a bond is broken on the p-side and the electron freed from the bond jumps to a broken bond on the n-side to complete it.

D-4. The depletion region of a P-N diode, under open circuit condition contains-
(1) Electrons (2) Holes
(3) Unmasked immobile impurity ions (4) Impurity atoms

D-5. Which is the wrong statement in following sentence? A device in which P and N type semiconductors are used is more useful then a vacuum tube because-
(1) power is not necesary to heat the filament
(2) it is more stable
(3) very less heat is producted in it
(4) its efficiency is high due to a high voltage drop across the junction

D-6. Depletion layer in P-N junction is caused by-
(1) Drift of holes
(2) Diffusion of free charge carriers
(3) Migration of impurity ions
(4) Drift of electrons

D-7. The contact potential at the junction site in a P-N junction is-
(1) positive on P side and negative on N side
(2) negative on P side and positive on N side
(3) zero
(4) infinite

D-8. When value of current increase in P-N junction, then the value of contact potential-
(1) decrease
(2) increase
(3) remain unchanged
(4) depends on temperature

D-9. What accounts for the flow of charge carriers in forward and reverse biasing of silicon P-N diode-
(1) Drift in both reverse and forward bias
(2) Drift in forward bias and diffusion in reverse bias
(3) Drift in reverse bias and diffusion in forward bias
(4) Diffusion in both forward and reverse bias

D-10. The barrier potential in a P-N junction is maximum in-
(1) the reverse bias condition
(2) the forward bias condition
(3) the condition when the junction diode is used as rectifier
(4) zero bias condition

D-11. The diffusion current in a P-N junction flows-
(1) from the N-side to the P-side
(2) from the P-side to the N-side
(3) from the N-side to the P-side if the junction is forward-biased and in the opposite direction if it is reverse baised
(4) from the P-side to the N-side if the junction is forward-biased and in the opposite direction if it is reverse biased

D-12. The drift current in a P-N junction flows -
(1) from the N-side to the P-side
(2) from the P-side to the N-side
(3) from the N-side to the P-side if the junction is forward-biased and in the oposite direction if it is reverse biased.
(4) from the P-side to the N-side if the junction is forward-baised and in the opposite direction if it is reverse baised.

D-13. For a reverse bias P-N junction-
(1) P region is positive and current is due to electrons
(2) P region is positive and the current is due to holes
(3) P region is negative and the current is due to electrons
(4) P region is negative and current is due to both electrons and holes

D-14. During P-N junction formation when the electron and holes stops moving from P to N and N to P, then
(1) There is increase in number of +ve and –ve ions at junction
(2) There is increase in number of electrons at junction
(3) There is increase in number of holes at junction
(4) There is increase in number of holes and electrons at junction

D-15. The value of barrier potential of P-N junction in Ge is-
(1) 0.03 volt in the direction of forward current
(2) 0.3 volt in the direction opposite of the forward current
(3) 25 volt in the direction opposite to the forward current
(4) 25 volt in the direction of the forward current

D-16. Region which have no free electron and holes in a P-N juction is-
(1) P-region (2) N-region (3) junction (4) depletion region

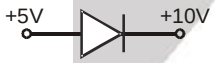
D-17. A device whose one end is connected to –ve terminal and other end connected to +ve terminal. If both ends are interchanged with supply then current is not flowing then device will be-
(1) P-N junction (2) Transistor (3) Zener diode (4) Triode

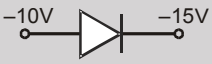
D-18. Potential barrier developed in a junction diode opposes-
(1) Minority carriers in both regions only (2) Majority carriers
(3) Electrons in N-region (4) Holes in P-region

D-19. The P-N junction diode works as insulator, if connected-
(1) To a.c. (2) In forward bias (3) In reverse bias (4) None of these

D-20. If the forward voltage in a diode is increased, the width of the depletion region-
(1) Decreases (2) Increases (3) Fluctuates (4) does not change

D-21. The resistance of a reverse baised P-N junction diode is about-
(1) 1 ohm (2) 10^2 ohm (3) 10^3 ohm (4) 10^6 ohm

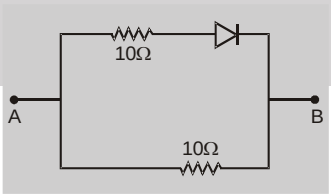
D-22. In which case is the juction diode is not reverse bias-
(1) 

(2) 

(3) 

(4) 

D-23. If V_A and V_B denote the potentials of A and B, then the equivalent resistance between A and B in the adjoint electric circuit is-

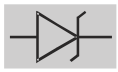
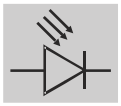
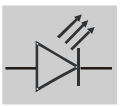


(1) 10 ohm if $V_A > V_B$ (2) 5 ohm if $V_A < V_B$
(3) 5 ohm if $V_A > V_B$ (4) 20 ohm if $V_A > V_B$

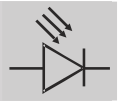
D-24. The avalanche breakdown in P-N junction is due to-
(1) Shift of Fermi level
(2) Cumulative effect of conduction band electron collision
(3) Widening of forbidden gap
(4) Low impurity concentration

D-25. If the two ends of a P-N junction are joined by a wire-
(1) There will not be a steady current in the circuit
(2) There will be a steady current from the N-side to the P-side
(3) There will a steady current from the P-side to the N-side
(4) There may or may not be a current depending upon the resistance of the connecting wire.

D-26. Symbolic representation of photodiode is-

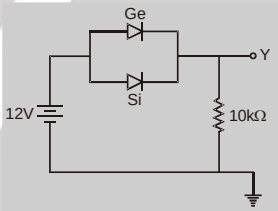
(1)  (2)  (3)  (4) 

D-27. Symbol of zener diode

(1)  (2)  (3)  (4) 

D-28. Consider the following statements 1 and 2 and identify the correct answer-
[1] A zener diode is always connected in reverse bias.
[2] The potential barrier of a P-N junction lies between 0.1 to 0.3 V approximately.
(1) [1] and [2] are correct (2) [1] and [2] are wrong
(3) [1] is correct, but [2] is wrong (4) [1] is wrong, but [2] is correct

D-29. Two junction diodes one of germanium (Ge) and other of silicon (Si) are connected as shown in figure to a battery of emf 12 V and a load resistance 10 kΩ. The germanium diode conducts at 0.3 V and silicon diode at 0.7 V. When a current flows in the circuit, the potential of terminal Y will be-



(1) 12 V (2) 11 V (3) 11.3 V (4) 11.7 V

D-30. When light falls on a photo diode, its conductivity increases. Experimentally it is found that the conductivity changes only when the wavelength of the incident light is less than 620 nm. The band gap of the diode is-
(1) 0.75 eV (2) 1.1 eV (3) 1.5 eV (4) 2.0 eV

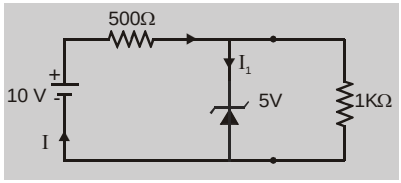
D-31. If the forward voltage in a semiconductor diode is changed form 0.5V to 2V, then the forward current changes by 1.5 mA. The forward resistance of diode will be-
(1) 1KΩ (2) 2KΩ (3) 4KΩ (4) 8KΩ

D-32. When the reverse potential in a semiconductor diode are 10V and 20V, then the corresponding reverse currents are 25μA and 50μA respectively. The reverse resistance of junction diode will be-
(1) 40Ω (2) $4 \times 10^5\ \Omega$ (3) 40KΩ (4) $4 \times 10^{-5}\ \Omega$

D-33. If the frequency of input alternating potential is n, then the ripple frequency of output potential of full wave rectifier will be-

(1) 2n (2) n (3) $\frac{n}{2}$ (4) $\frac{n}{4}$

D-34. The current flowing through the zener diode in fig. is-



(1) 20 mA (2) 25 mA (3) 15 mA (4) 5 mA

D-35. Function of rectifier is

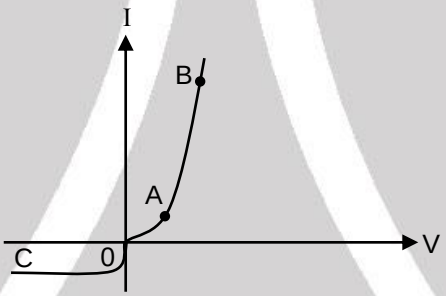
(1) To convert ac into dc (2) To convert dc into ac

(3) Both (1) and (2) (4) None of these

D-36. The electrical circuits used to get smooth d.c. output from a rectified circuit is called

(1) oscillator (2) full wave rectifier (3) amplifier (4) filter

D-37. The adjacent figure shows I – V characteristics of a silicon diode. In this connection three statements are made – (I) the region OC corresponds to reverse bias of the diode, (II) the voltage at point A is about 0.2 volt, and (III) different scales have been used along +ve and –ve directions of Y-axis. Therefore,



(1) only statement (I) is correct

(2) only statements (I) and (II) are correct

(3) only statements (I) and (III) are correct

(4) all statements (I),(II) and (III) are correct

Section (E) : Transistors

E-1. In a transistor

(1) The emitter has the least concentration of impurity

(2) The collector has the least concentration of impurity

(3) The base has the least concentration of impurity

(4) All the three region have equal concentration of impurity

E-2. The thinest part of a transistor is-

(1) emitter (2) base

(3) collector (4) according to transistor parameters none of these

E-3. In transistor symbols, the arrows shows the direction of-

(1) current in the emitter (2) electron current in the emitter

(2) holes current in the emitter (4) none of these

E-4. Input resistance of transistor in comparision to output resistance is (Amplifier)-

(1) Low (2) High (3) Low and high (4) None of these

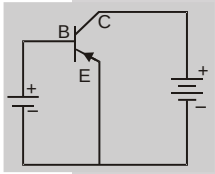
E-5. In a normal operation of a tranistor,
(1) the base-emitter junction is forward-biased (2) the base-collector junction is forward-biased
(3) the base-emitter junction is reverse-baised (4) the base-collector junction is reverse-baised

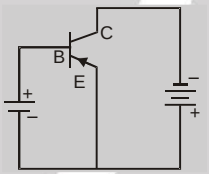
E-6. In a properly biased transistor-
(1) Both depletion layers are equally large.
(2) Both depletion layers are equally small.
(3) Emitter-base depletion layer is large but base collector depletion layer is small.
(4) Emitter-base depletion layer is small but base-collector depletion layer is large.

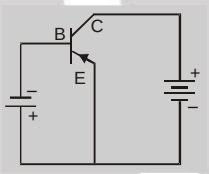
E-7. In an N-P-N transistor, the emitter current is-
(1) Slightly more than the collector current (2) Slightly less than the collector current
(3) Equal to the collector current (4) Equal to the base current

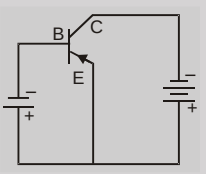
E-8. A N-P-N transistor conducts when-
(1) Both collector and emitter are positive with respect to the base.
(2) Collector is positive and emitter is negative with respect to the base.
(3) Collector is positive and emitter is at same potential as the base.
(4) Both collector and emitter are negative with respect to the base.

E-9. Which one of the following circuits shows correct biasing of a PNP transistor to operate in active region in the CE mode-

(1) 

(2) 

(3) 

(4) 

E-10. To use trasistor as an amplifier-
(1) emitter base junction is in forward biased and collector junction is reverse biased.
(2) biasing voltage are not required
(3) both junction are forward biased
(4) both junction are reverse biased

E-11. An amplifier is nothing but an oscillator with –
(1) positive feedback (2) high gain (3) no feed back (4) negative feed back

E-12. When N-P-N transistor is used as an amplifier-
(1) Electrons move from base to emitter (2) Electrons move from emitter to base
(3) Electrons move from collector to base (4) Holes move from base to emitter

E-13. A common emitter circuit is used as an amplifier, its current gain is 50. If input resistance is 1 kΩ and input voltage is 5 volt then output current will be-
(1) 250 mA (2) 30 mA (3) 50 mA (4) 100 mA

E-14. In a N-P-N transistor circuit, the collector current is 10 mA. If 90% of the electrons emitted reach the collector, the emiitter current (I_E) and base current (I_B) are given by-
(1) I_E = 1mA; I_B = 11 mA (2) I_E = 11 mA; I_B = 1 mA
(3) I_E = –1 mA; I_B = 9 mA (4) I_E = 9 mA ; I_B = –1 mA

E-15. Which of the following is true-
(1) Common base transistor is commonly used because current gain is maximum.
(2) Common-emitter is commonly used because current gain is maximum.
(3) Common collector is commonly used because current gain is maximum.
(4) Common emitter is the least used transistor.

E-16. For transistor relation in current amplification factors is-

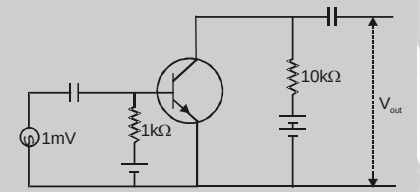
(1) $\alpha = \frac{\beta}{1-\beta}$ (2) $\beta = \frac{\alpha}{1-\alpha}$ (3) $\alpha = \frac{1+\beta}{\beta}$ (4) $\beta = \frac{\alpha}{1+\alpha}$

E-17. For a transistor, $\alpha = 0.9$, the value of β is-
(1) 1 (2) 0.09 (3) 0.9 (4) 9

E-18. In a P-N-P transistor working as a common-base amplifier current gain is 0.96 and emitter current is 7.2 mA The base current is-
(1) 0.4 mA (2) 0.2 mA (3) 0.29 mA (4) 0.35 mA

E-19. In an NPN transistor 10^{10} electrons emitted from the emitter in 10^{-6} s and 2% electrons recombine with holes in base, then current gain α and β are-
(1) 0.49, 9.8 (2) 0.58, 4.9 (3) 0.78, 49 (4) 0.98, 49

E-20. In the fig. a common emitter configuration on N-P-N transistor with current gain $\beta = 100$ is used. The output voltage of the amplifier will be-



(1) 10 mV (2) 0.1 V (3) 1.0 V (4) 10 V

E-21. A N-P-N transistor is connected in common emitter configuration in which collector supply is 8 volt and the voltage drop across the load resistor of 800 ohm connected to the collector circuit is 0.8 volt. If current amplification factor α is $\frac{25}{26}$ and the input resistance of the transistor is 200 ohm then the collector emitter voltage, base current, the voltage and power gain are-
(1) 3.5 V, 2×10^{-5} A and $A_V = 50$V, $A_P = 6500$ (2) 7.2 V, 4×10^{-5} A and $A_V = 100$V, $A_P = 2500$
(3) 4.5 V, 3×10^{-5} A and $A_V = 50$V, $A_P = 6500$ (4) 5.6 V, 3×10^{-5} A and $A_V = 60$V, $A_P = 7500$

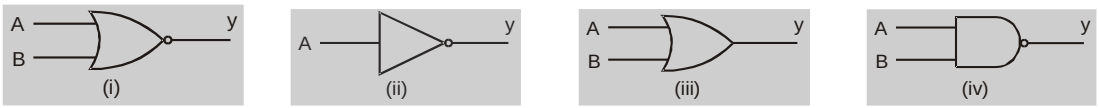
E-22. In an NPN transistor the values of base current and collector current are 100μA and 9 mA respectively, the emitter current will be-
(1) 9.1mA (2) 18.2mA (3) 9.1μA (4) 18.2μA

E-23. In a common emitter circuit, if V_{CE} is changed by 0.2V, then collector current changed by 4×10^{-3} mA. Output resistance will be-
(1) 10 kΩ (2) 30 kΩ (3) 50 kΩ (4) 70 kΩ

E-24. The following statement is not true-
(1) The oscillations in an oscillator are maintained.
(2) Oscillator is an active circuit.
(3) There is no voltage gain in an oscillator.
(4) Oscillator works as an amplifier having infinite voltage gain.

Section (F) : Logic Gates

F-1. Given below are four logic gates symbols. NAND, NOR and OR are respectively-



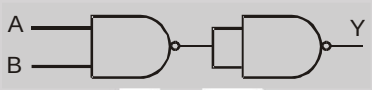
(1) (iv), (i), (iii) (2) (i), (iii), (iv) (3) (i), (ii), (iii) (4) (i), (iv), (ii)

F-2. The output of OR gate is 1-

(1) if both inputs are zero (2) if either or both inputs are 1

(3) only if both inputs are 1 (4) if either input is zero

F-3. Following diagram performs the logic function of-



(1) AND gate (2) NAND gate (3) OR gate (4) XOR gate

F-4. Given truth table is for which GATE -

A	B	Y
1	1	1
1	0	0
0	1	0
0	0	0

(1) AND (2) OR (3) NOR (4) NAND

F-5. NAND gate is-

(1) a basic gate (2) not a universal gate

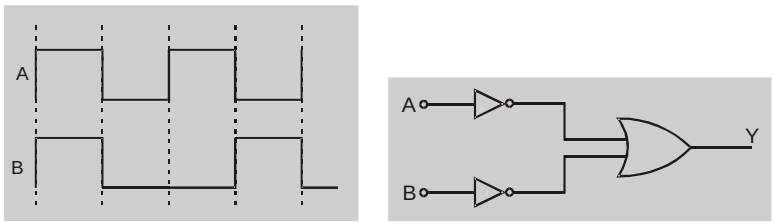
(3) a universal gate (4) none of these.

F-6. The given truth table is for which logic gate-

A	**B**	**Y**
1	1	0
0	1	1
1	0	1
0	0	1

(1) NAND (2) XOR (3) NOR (4) OR

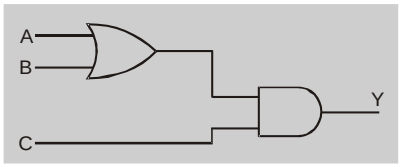
F-7. In a given circuit as shown the two inputs waveform A and B applied simultaneously.



The resultant wave form Y is-

(1)  (2)  (3)  (4) 

F-8. For the logic circuit shown the Boolean relation is-



(1) Y = ABC (2) Y = A + BC (3) Y = (A + B) C (4) Y = AB + C

F-9. For the given combination of gates, if the logic states of inputs A, B, C are as follows A = B = C = 0 and A = B = 1, C = 0 then the logic states of output D are



(1) 0, 0 (2) 0, 1 (3) 1, 0 (4) 1, 1

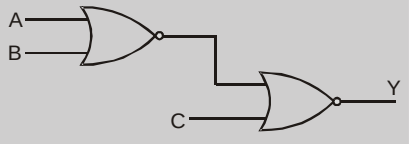
F-10. The Boolean equation of NOR gate is-

(1) C = A + B (2) $C = \overline{A+B}$ (3) C = A + B (4) $C = \overline{A \cdot B}$

F-11. What will be the input of A and B for the Boolean expression $(\overline{A+B}) \cdot (\overline{A \cdot B}) = 1$.

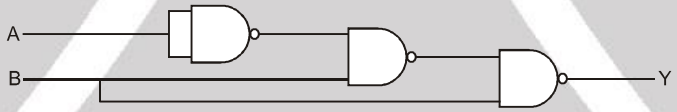
(1) 0, 0 (2) 0, 1 (3) 1, 0 (4) 1, 1

F-12. The Boolean expression for the output Y of the logic operation shown, is-



(1) $(\overline{A+B})+C$ (2) $(A+B)+\overline{C}$ (3) $\overline{(A+B)+C}$ (4) $(\overline{A+B+C})$

F-13. The arrangement of NAND gates shown below effectively works as



(1) AND gate (2) OR gate (3) NAND gate (4) NOR gate

Exercise-2

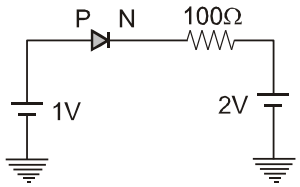
Marked Questions can be used as Revision Questions.

PART - I : OBJECTIVE QUESTIONS

1. The depletion layer in silicon diode is 1 μm wide and the knee potential is 0.6 V, then the electric field in the depletion layer will be

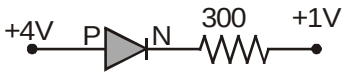
(1) Zero (2) 0.6 Vm^{-1} (3) 6×10^4 V/m (4) 6×10^5 V/m

2. The current through an ideal PN-junction shown in the following circuit diagram will be



(1) Zero (2) 1 mA (3) 10 mA (4) 30 mA

3. In the circuit given below, the value of the current is

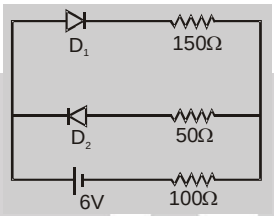


(1) 0 amp (2) 10^{-2} amp (3) 1 amp (4) 0.10 amp

4. The maximum efficiency of full wave rectifier is

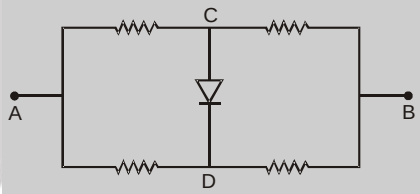
(1) 100 % (2) 25.20 % (3) 40.2 % (4) 81.2 %

5. The circuit shown in the figure contains two diodes each with a forward resistance of 50 ohm and with infinite reverse resistance. If the battery voltage is 6V, find the current through the 100 ohm resistance-



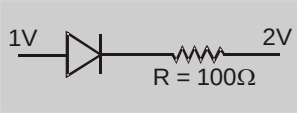
(1) 0.01 A (2) 0.05 A (3) 0.02 A (4) 0.03 A

6. Four equal resistors, each of resistance 10 ohm are connected as shown in the adjoining circuit diagram. Then the equivalent resistance between points A and B is-



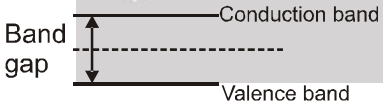
(1) 40 ohm (2) 20 ohm (3) 10 ohm (4) 5 ohm

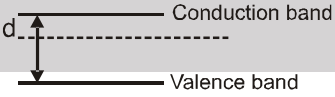
7. In the arrangement shown in fig. the current through diode is-

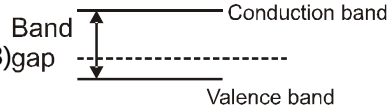


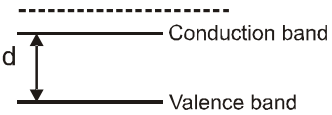
(1) 10 mA (2) 1 mA (3) 20 mA (4) Zero

8. Which one of the following diagrams correctly represents the energy levels in the p–type semiconductor?

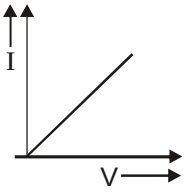
(1) 

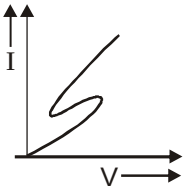
(2) 

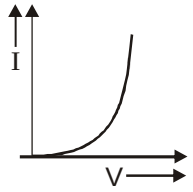
(3) 

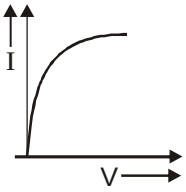
(4) 

9. Choose the correct option for the forward biased characteristics of a p–n junction.

(1) 

(2) 

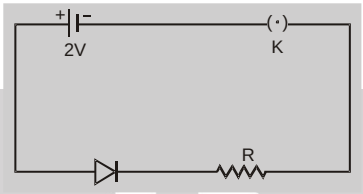
(3) 

(4) 

10. In the case of constants α and β of a transistor
(1) $\alpha = \beta$ (2) $\beta < 1$, $\alpha > 1$ (3) $\alpha\beta = 1$ (4) $\beta > 1$, $\alpha < 1$

11. If $\alpha = 0.98$ and current through emitter $i_e = 20$ mA, the value of β is
(1) 4.9 (2) 49 (3) 96 (4) 9.6

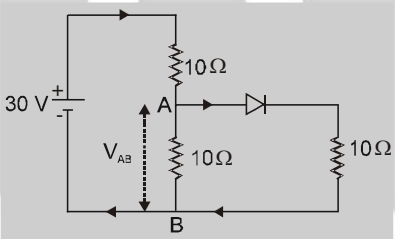
12. A diode made forward biased by a two volt battery however there is a drop of 0.5 V across the diode which is independent of current. Also a current greater then 10 mA produces large joule loss and damages diode. If diode is to be operated at 5 mA, the series resistance to be put is-



(1) 3 KΩ (2) 300 KΩ (3) 300 Ω (4) 200 KΩ

13. The ratio of resistance for forward to reverse bias of P–N junction diode is-
(1) $10^2 : 1$ (2) $10^{-2} : 1$ (3) $1 : 10^{-4}$ (4) $1 : 10^4$

14. For given circuit potential difference V_{AB} is

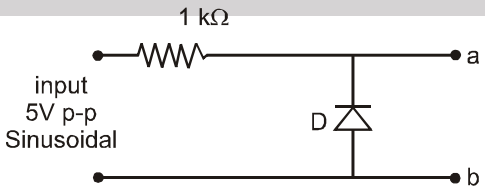


(1) 10 V (2) 20 V (3) 30 V (4) None

15. Zener diode is used-
(1) As an amplifier (2) As a rectifier
(3) As an oscillator (4) As a voltage regulator

16. Zener breakdown will occur if-
(1) Impurity level is low (2) Impurity level is high
(3) Impurity is less in n-side (4) Impurity is less in p-side

17.* Refer to the circuit given below. Output voltage V_0 is measured between points a and b. Then,



(1) the peak value of V_0 is 2.5 volt above the minimum if the diode is assumed to be ideal.
(2) the positive half cycle of the input is clipped.
(3) the circuit acts as a rectifier.
(4) the peak value of V_0 is about 3.2 volt above the minimum if D is silicon diode (non-ideal).

18. An ac source (sinusoidal source with frequency 50 Hz) is connected in series with a rectifying diode, a 100 Ω resistor, a 1000 μF capacitor and milliammeter. After some time the milliammeter reads zero. The voltage measured across the capacitor with a dc voltmeter is
(1) the peak voltage of the ac source
(2) rms voltage of the ac source
(3) average voltage of the ac source over a half cycle
(4) average voltage of the ac source over a full cycle.

19. The fundamental frequency of the output of a bridge rectifier driven by ac mains is
(1) 50 Hz (2) zero (3) 100 Hz (4) 25 Hz

20. Which one of the following devices does not respond to the intensity of light incident on it ?
(1) Photoresistor (LDR) (2) Photodiode
(3) Light Emitting Diode (4) Solar Cell

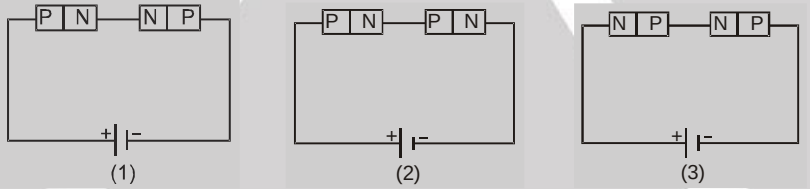
21.*_ In a bipolar junction transistor
(1) the most heavily doped region is the emitter
(2) the level of doping is the same in both the emitter and the collector
(3) its base is the thinnest part
(4) when connected in common emitter configuration a base current is generally of the order of μA

Exercise-3

✍ **Marked Questions may have for Revision Questions.**

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

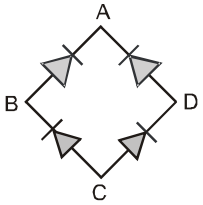
1. Two identical P-N junction may be connected in series with a battery in three ways (fig below). The potential drops across the two P-N junction are equal in-



(1) Circuit 1 and 2 (2) Circuit 2 and 3 (3) Circuit 3 and 1 (4) Circuit 1 only

2. In a P-N junction diode which is not connected to any circuit-
(1) Potential is the same every where
(2) The P-type side is at a higher potential than the N-type side
(3) There is an electric field at the junction directed from the N-type side to the P-type side
(4) There is an electric field at the junction directed from the P-type side to the N-type side

3.✍ For the given circuit shown in fig, to act as full wave rectifier, a.c. input should be connected across .....and.....the d.c. output would appear across.....and.....



(1) A, C and B, D (2) B, D and A, C (3) A, B and C, D (4) C, A and D, B

4. In a.....biased P-N junction the net flow holes is from N-region to the P-region-
(1) forward (2) reverse (3) no (4) both 1 and 2

PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. At absolute zero, Si acts as : **[AIEEE-2002]**
(1) non-meta (2) metal
(3) insulator (4) none of these

2. By increasing the temperature, the specific resistance of a conductor and semiconductor: **[AIEEE-2002]**
(1) increases for both
(2) decreases for both
(3) increases, decreases respectively
(4) decreases, increases respectively

3. The energy band gap is maximum in : **[AIEEE-2002]**
(1) metals (2) superconductors
(3) insulators (4) semiconductors

4. The part of a transistor which is most heavily doped to produce large number of majority charge carries is : **[AIEEE-2002]**
(1) emitter (2) base
(3) collector (4) can be any of the above three

5. A strip of copper and another of germanium are cooled from room temperature to 80 K. The resistance of : **[AIEEE-2003]**
(1) each of these decreases
(2) copper strip increases and that of germanium decreases
(3) copper strip decreases and that of germanium increases
(4) each of these increases

6. The difference in the variation of resistance with temperature in a metal and a semiconductor arises essentially due to the difference in the : **[AIEEE-2003]**
(1) crystal structure
(2) variation of the number of charge carries with temperature
(3) type of bonding
(4) variation of scattering mechanism with temperature

7. When P-N junction diode is forward biased, then- **[AIEEE-2004]**
(1) the depletion region is reduced and barrier height is increased.
(2) the depletion region is widened and barrier height is reduced.
(3) both the depletion region and barrier height are reduced.
(4) both the depletion region and barrier height are increased.

8. When npn transistor is used as an amplifier: **[AIEEE-2004]**
(1) electrons move from base to collector
(2) holes move from emitter to base
(3) electrons move from collector to base
(4) holes move from base to emitter

9. The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than 2480 nm, is incident on it. The band gap in (eV) for the semiconductor is : **[AIEEE-2005]**

(1) 1.1 eV (2) 2.5 eV (3) 0.5 eV (4) 0.7 eV

10. In a common base amplifier, the phase difference between the input signal voltage and output voltage is : **[AIEEE-2005]**

(1) $\frac{\pi}{4}$ (2) π (3) zero (4) $\frac{\pi}{2}$

11. In a full wave rectifier circuit operating from 50 Hz mains frequency, the fundamental frequency in the ripple would be : **[AIEEE-2005]**

(1) 50 Hz (2) 25 Hz (3) 100 Hz (4) 70.7 Hz

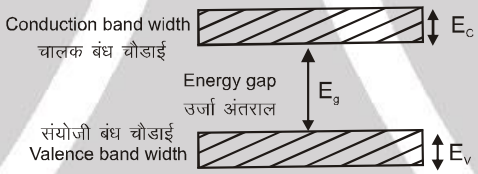
12. If the ratio of the concentration of electrons to that of holes in a semiconductor is $\frac{7}{5}$ and the ratio of currents is $\frac{7}{4}$, then what is the ratio of their drift velocities ? **[AIEEE 2006]**

(1) 5/8 (2) 4/5 (3) 5/4 (4) 4/7

13. In a common–base of mode of transistor, the collector current is 5.488 mA for an emitter current of 5.60 mA. The value of the base current amplification factor (β) will be : **[AIEEE 2006]**

(1) 49 (2) 50 (3) 51 (4) 48

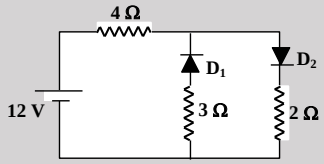
14. If the lattice constant of this semiconductor is decreased, then which of the following is correct? **[AIEEE 2006]**



(1) All E_C, E_g , E_v increase (2) E_C and E_V increase, but E_g decreases

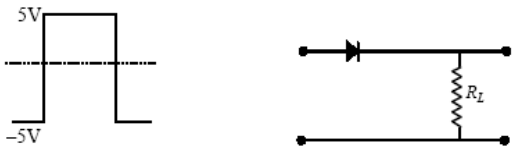
(3) E_c and E_v decrease, but E_g increases (4) All E_C,E_g , E_v decrease

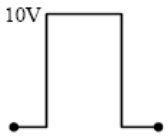
15. The circuit has two oppositely connected ideal diodes in parallel. What is the current flowing in the circuit? **[AIEEE 2006]**

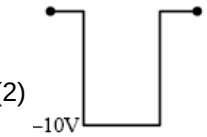


(1) 2.31 A (2) 1.33 A (3) 1.71 A (4) 2.00 A

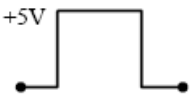
16. If in p-n junction diode, a square input signal of 10 V is applied as shown. Then the output signal across R_L will be **[AIEEE 2007]**



(1) 

(2) 

(3) 

(4) 

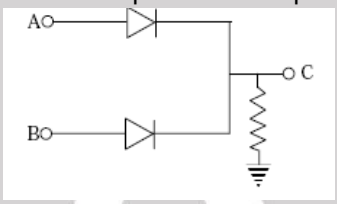
17. Carbon, silicon and germanium have four valence electrons each. At room temperature which one of the following statements is most appropriate? **[AIEEE 2007]**

(1) the number of free conduction electrons is significant in *C* but small in *Si* and *Ge*.
(2) the number of free conduction electrons is negligibly small in all the three.
(3) the number of free electrons for conduction is significant in all the three.
(4) the number of free electrons for conduction is significant only in *Si* and *Ge* but small in *C*.

18. A working transistor with its three legs marked P, Q and R is tested using a multimeter. No conduction is found between P and Q. By connecting the common (negative) terminal of the multimeter to R and the other (positive) terminal to P or Q, some resistance is seen on the multimeter. Which of following is true for the transistor ? [**AIEEE 2008]**

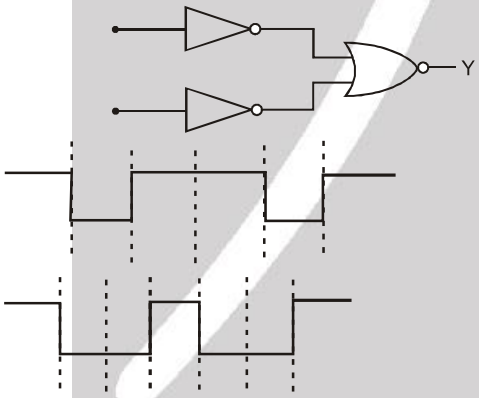
(1) It is a pnp transistor with R as collector (2) It is a pnp transistor with R as emitter
(3) It is an npn transistor with R as collector (4) It is an npn transistor with R as base

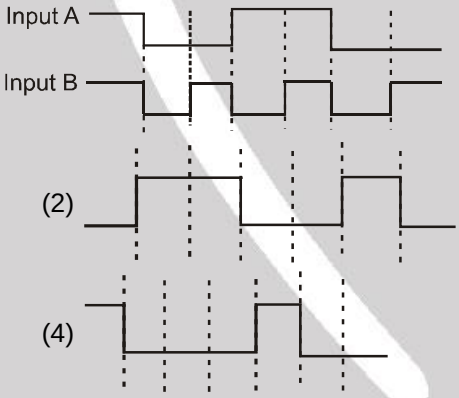
19. In the circuit below, A and B represent two inputs and C represents the output. **[AIEEE 2008]**



The circuit represents
(1) AND gate (2) NAND gate (3) OR gate (4) NOR gate

20. The logic circuit shown below has the input waveforms 'A' and 'B' as shown. Pick out the correct out put waveform. **[AIEEE 2009]**





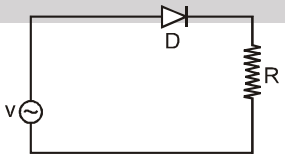
(1)

(2)

(3)

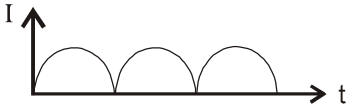
(4)

21. A p-n junction (4) shown in the figure can act as a rectifier. An alternating current source (V) is connected in the circuit. **[AIEEE 2009]**

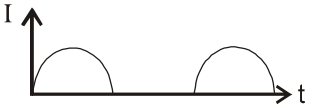


The current (I) in the resistor (R) can be shown by:

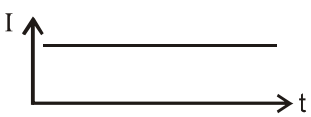
(1)



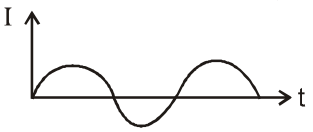
(2)



(3)



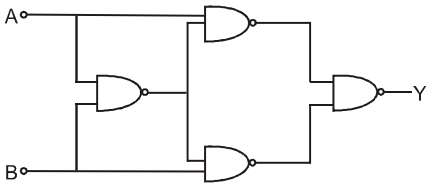
(4)



22. The output of an OR gate is connected to both the inputs of a NAND gate. The combination will serve as a : **[AIEEE 2011, 11 May; 4, –1]**

(1) NOT gate (2) NOR gate (3) AND gate (4) OR gate

23. Truth table for system of four NAND gates as shown in figure is : **[AIEEE 2012 ; 4, –1]**



(1)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(2)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(3)

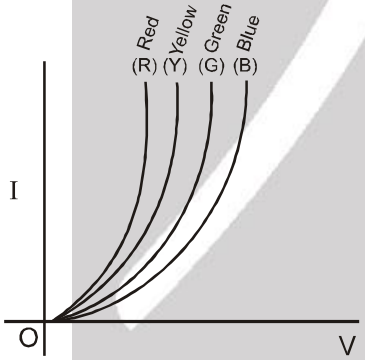
A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(4)

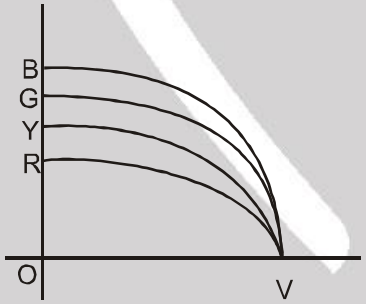
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

24. The I – V characteristic of an LED is **[JEE (Main) 2013, 4/120, –1]**

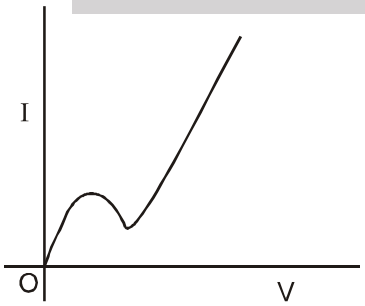
(1)



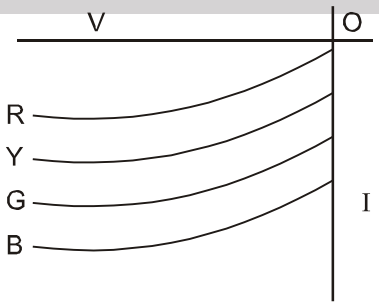
(2)



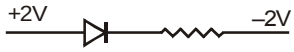
(3)

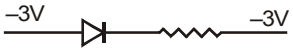


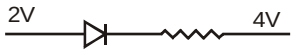
(4)

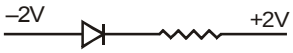


25. The forward biased diode connection is **[JEE-MAIN 2014 ; 4/120. –1]**

(1) 

(2) 

(3) 

(4) 

26. The temperature dependence of resistances of Cu and undoped Si in the temperature range 300 – 400 K, is best described by : **[JEE-MAIN 2016 ; 4/120. –1]**

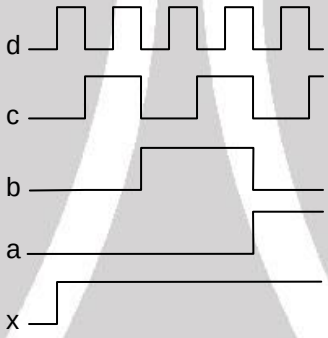
(1) Linear increase for Cu, exponential increase for Si

(2) Linear increase for Cu, exponential decrease for Si

(3) Linear decrease for Cu, linear decrease for Si

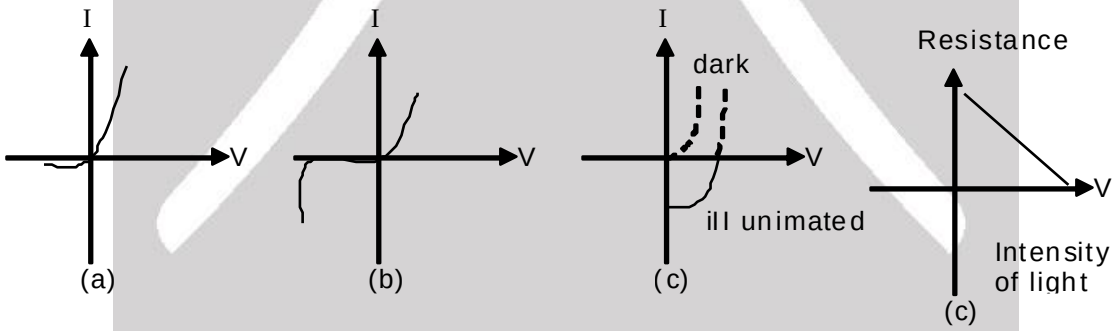
(4) Linear increase for Cu, linear increase for Si

27. If a,b,c,d are inputs to a gate and x is its output, then, as per the following time graph, the gate is : **[JEE-MAIN 2016 ; 4/120. –1]**



(1) AND (2) OR (3) NAND (4) NOT

28. Identify the semiconductor devices whose characteristics are given below, in the order (1),(2),(3),(4)



(1) zener diode, simple diode, Light dependent resistance, Solar cell

(2) Solar cell , Light dependent resistance, Zener diode, simple diode

(3) Zener diode, Solar cell, Simple diode, Light dependent resistance

(4) Simple diode, Zener diode, Solar cell, Light dependent resistance.

29. For a common emitter configuration, if α and β have their usual meanings, the **incorrect** relationship between α and β is. **[JEE (Main) 2016 ; 4/120, –1]**

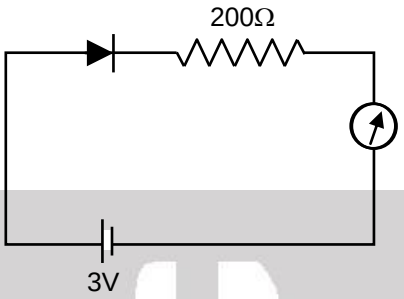
(1) $\alpha = \frac{\beta}{1-\beta}$ (2) $\alpha = \frac{\beta}{1+\beta}$ (3) $\alpha = \frac{\beta^2}{1+\beta^2}$ (4) $\frac{1}{\alpha} = \frac{1}{\beta} + 1$

30. In a common emitter amplifier circuit using an n-p-n transistor, the phase difference between the input and the output voltages will be: **[JEE (Main) 2017; 4/120, –1]**

(1) 180° (2) 45° (3) 90° (4) 135°

31. The reading of the ammeter for a silicon diode in the given circuit is :

[JEE (Main) 2018, 4/120, –1]



(1) 11.5mA (2) 13.5 mA (3) 0 (4) 15 mA

Answers

EXERCISE – 1

Section (A)

A-1.	(2)	**A-2.**	(3)	**A-3.**	(1)
A-4.	(4)	**A-5.**	(1)	**A-6.**	(4)
A-7.	(3)	**A-8.**	(2)	**A-9.**	(4)

Section (B)

B-1.	(2)	**B-2.**	(1,2,3,4)

Section (C)

C-1.	(3)	**C-2.**	(4)	**C-3.**	(3)
C-4.	(3)	**C-5.**	(2)	**C-6.**	(2)
C-7.	(4)	**C-8.**	(2)	**C-9.**	(1)
C-10.	(2)	**C-11.**	(1)	**C-12.**	(3)
C-13.	(1)	**C-14.**	(1)	**C-15.**	(1)

Section (D)

D-1.	(2)	**D-2.**	(1)	**D-3.**	(3)
D-4.	(3)	**D-5.**	(4)	**D-6.**	(2)
D-7.	(2)	**D-8.**	(1)	**D-9.**	(3)
D-10.	(1)	**D-11.**	(2)	**D-12.**	(1)
D-13.	(4)	**D-14.**	(1)	**D-15.**	(2)
D-16.	(4)	**D-17.**	(1)	**D-18.**	(2)
D-19.	(3)	**D-20.**	(1)	**D-21.**	(4)
D-22.	(2)	**D-23.**	(3)	**D-24.**	(2)
D-25.	(1)	**D-26.**	(3)	**D-27.**	(1)
D-28.	(3)	**D-29.**	(4)	**D-30.**	(4)
D-31.	(1)	**D-32.**	(2)	**D-33.**	(1)
D-34.	(4)	**D-35.**	(1)	**D-36.**	(4)
D-37.	(3)				

Section (E)

E-1.	(3)	**E-2.**	(2)	**E-3.**	(1)
E-4.	(1)	**E-5.**	(1, 4)	**E-6.**	(4)
E-7.	(1)	**E-8.**	(2)	**E-9.**	(4)
E-10.	(1)	**E-11.**	(1)	**E-12.**	(2)
E-13.	(1)	**E-14.**	(2)	**E-15.**	(2)
E-16.	(2)	**E-17.**	(4)	**E-18.**	(3)
E-19.	(4)	**E-20.**	(3)	**E-21.**	(2)
E-22.	(1)	**E-23.**	(3)	**E-24.**	(3)

Section (F)

F-1.	(1)	**F-2.**	(2)	**F-3.**	(1)
F-4.	(1)	**F-5.**	(3)	**F-6.**	(1)
F-7.	(1)	**F-8.**	(3)	**F-9.**	(4)
F-10.	(2)	**F-11.**	(1)	**F-12.**	(3)
F-13.	(3)				

EXERCISE – 2

PART– I

1.	(4)	**2.**	(1)	**3.**	(2)
4.	(4)	**5.**	(3)	**6.**	(3)
7.	(4)	**8.**	(3)	**9.**	(3)
10.	(4)	**11.**	(2)	**12.**	(3)
13.	(4)	**14.**	(1)	**15.**	(4)
16.	(2)	**17.**	(1,2,3,4)	**18.**	(3)
19.	(3)	**20.**	(3)	**21.**	(1,3,4)

EXERCISE – 3

PART – I

1.	(2)	**2.**	(3)	**3.**	(2)
4.	(2)				

PART – II

1.	(3)	**2.**	(3)	**3.**	(3)
4.	(1)	**5.**	(3)	**6.**	(2)
7.	(3)	**8.**	(4)	**9.**	(3)
10.	(3)	**11.**	(3)	**12.**	(3)
13.	(1)	**14.**	(2)	**15.**	(4)
16.	(4)	**17.**	(4)	**18.**	(4)
19.	(3)	**20.**	(4)	**21.**	(2)
22.	(2)	**23.**	(1)	**24.**	(1)
25.	(1)	**26.**	(2)	**27.**	(2)
28.	(4)	**29.**	(1,3)	**30.**	(1)
31.	(1)				